

COVID-19 Infection Segmentation

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I. EXPLORATORY DATA ANALYSIS

A. Dataset Overview

The research uses the COVID-QU-Ex dataset, which was created by researchers at Qatar University. This is currently the largest lung mask dataset available for medical studies. It contains a total of 33,920 chest X-ray images, including 11,956 COVID-19 cases, 11,263 non-COVID infections like pneumonia, and 10,701 normal cases.

B. Training and Test Sets

The experiments are conducted on two different data groups, each divided into training, validation, and test sets:

- Lung Segmentation Data: This includes all 33,920 images with their corresponding lung masks.
- Infection Segmentation Data: This is a smaller subset of 5,826 images. It includes COVID-19 samples with both lung and infection masks, while the normal and non-COVID samples provide lung masks.

C. Class Distribution

The tables below show how the images are distributed across different categories and sets.

TABLE I
DATA DISTRIBUTION OF THE INFECTION SEGMENTATION SUBSET

Split	Class	Images	Lung Masks	Infection Masks
Train	COVID-19	1864	1864	1864
Train	Non-COVID	932	932	932
Train	Normal	932	932	932
Validation	COVID-19	466	466	466
Validation	Non-COVID	233	233	233
Validation	Normal	233	233	233
Test	COVID-19	583	583	583
Test	Non-COVID	292	292	292
Test	Normal	291	291	291
Total	COVID-19	2913	2913	2913
Total	Non-COVID	1457	1457	1457
Total	Normal	1456	1456	1456

The infection segmentation subset has a balanced partition. This ensures the model is evaluated fairly and helps reduce any potential bias during training.

TABLE II
DATA DISTRIBUTION OF THE LUNG SEGMENTATION DATASET

Split	Class	Images	Lung Masks
Train	COVID-19	7658	7658
Train	Non-COVID	7208	7208
Train	Normal	6849	6849
Validation	COVID-19	1903	1903
Validation	Non-COVID	1802	1802
Validation	Normal	1712	1712
Test	COVID-19	2395	2395
Test	Non-COVID	2253	2253
Test	Normal	2140	2140
Total	COVID-19	11956	11956
Total	Non-COVID	11263	11263
Total	Normal	10701	10701
Total	All	33920	33920

II. METHODOLOGY

The proposed method follow a two-stage segmentation process to accurately identify lung areas and COVID-19 infection regions from chest X-ray images. The pipeline is built using the PyTorch framework

A. Preprocessing and Data Augmentation

All images and masks are resized to 128x128 pixels to reduce computational cost. To improve the model's ability, some data augmentation techniques are applied during the training phase. These include horizontal flipping, random rotation, and brightness adjustments. All pixel values are normalized to a range of [0, 1].

B. Model Architecture: TinyUNet

The model is specifically designed to operate under limited computational resources, because the training process was by google colab. To reduce model complexity, TinyUNet replaces standard convolutions with depthwise separable convolutions. The model contains approximately 9.9×10^4 parameters,

The network follows the classical encoder-decoder structure of U-Net. The encoder path extracts hierarchical contextual features, while the symmetric decoder path progressively recovers spatial resolution for accurate pixel-level segmentation. Skip connections are employed between corresponding encoder and decoder layers to retain fine-grained spatial information and improve localization accuracy.

C. Two-Stage Segmentation Strategy

The segmentation process is divided into two distinct stages:

- Stage A (Lung Segmentation): In this stage, the model is trained to identify the boundary of the lungs from the original X-ray images. This creates a lung mask that filters out background information like bones or other organs.
- Stage B (Infection Segmentation): This stage focuses on identifying the specific areas of infection. The input for this stage is the original image multiplied by the lung mask (Region of Interest - ROI). By focusing only on the lung area, the model can more accurately detect COVID-19 infections.

D. Training and Evaluation

The models are trained using a combination of Binary Cross-Entropy (BCE) and Dice loss functions. The Adam optimizer is used with a learning rate of 0.001. To evaluate the performance, the Dice coefficient and Intersection-over-Union (IoU) metrics are calculated on the test set.

III. RESULTS

A. Stage A: Lung Segmentation

Stage A demonstrated strong performance in lung boundary localization. On the test set, the model achieved a Dice score of 0.9683 and an IoU of 0.9406. The quantitative results for Stage A are summarized in Table III.

TABLE III
PERFORMANCE METRICS FOR STAGE A

Loss (BCE)	Dice	IoU
0.0360	0.9683	0.9406

B. Stage B: Infection Segmentation

The proposed model achieved promising performance on the test set, with a Dice score of 0.8277 and an IoU of 0.7689.

Compared to Stage A, the higher loss value reflects the increased difficulty of accurately segmenting infection regions. Overall, these results demonstrate the effectiveness of the two-stage framework, where precise lung localization in Stage A supports improved infection segmentation in Stage B.

The test performance for Stage B is reported in Table IV.

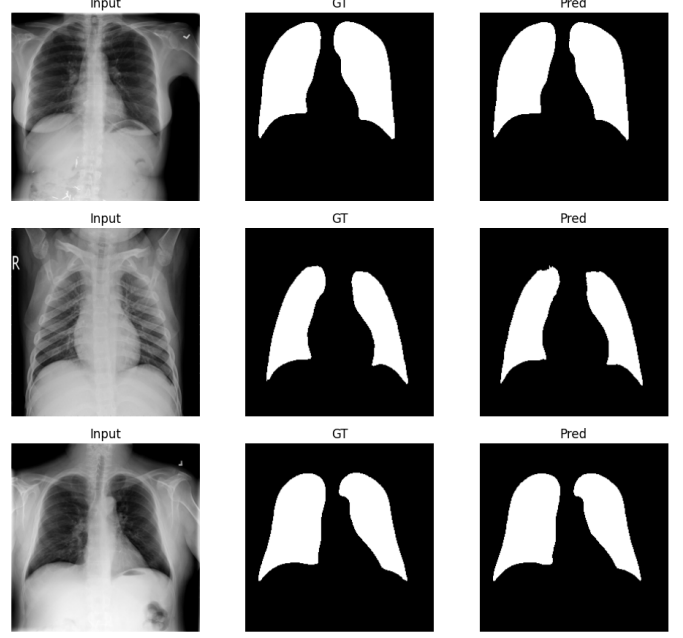


Fig. 1. Qualitative lung segmentation results on the test set.

TABLE IV
PERFORMANCE METRICS FOR STAGE B (INFECTION SEGMENTATION) ON THE TEST SET

Loss (BCE)	Dice	IoU
0.0621	0.8277	0.7689

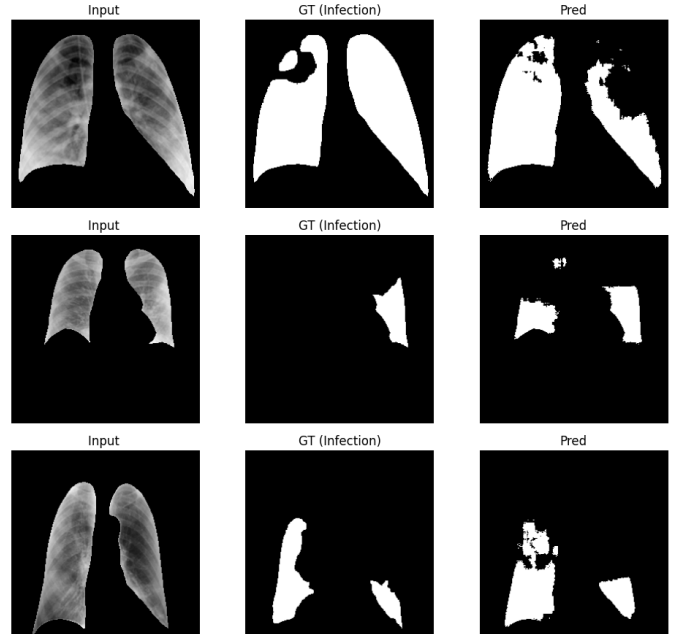


Fig. 2. Qualitative infection segmentation results on the test set.

C. Comparison

Table V summarizes representative segmentation performances reported in the literature for COVID-19 lung and infection segmentation. Relevant metrics such as Dice coefficient and IoU are used for comparison with the proposed TinyUNet.

TABLE V
COMPARISON WITH REPRESENTATIVE SEGMENTATION METHODS

Method	Modality	Dice	IoU
Attention U-Net	CT	0.784	0.827
3D U-Net	CT	0.761	–
U-Net (2D baseline)	CXR	–	0.748
TinyUNet	CXR	0.828	0.769

IV. CONCLUSION

Because of limited computational resources and the use of a compact model design, the segmentation accuracy for infection regions remains lower than that of larger state-of-the-art architectures. Despite this limitation, the proposed approach provides a reasonable trade-off between performance and efficiency, making it suitable for practical scenarios where computational resources are constrained.